

Modelling of a MMC HVDC Link between Koeberg Power Station and Cape Town - Experiences in simulation

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Abstract— This paper reports on a feasibility study of an HVDC link between Koeberg and Cape Town, South Africa, in light of the available technologies whilst noting new technological developments in the HVDC industry. The design is considered with special focus on power system stability and control with regards to future potential integration of renewable micro generators and smart grid technology into the western grid. Experiences are reported in terms of simulation of the line using Simplorer with an emphasis on building a model of a modular multi-level converter (MMC) in a commercial power electronics simulator.

Keywords—HVDC, Modular Multilevel Converter, simulation

I. INTRODUCTION

Cape Town and its neighbouring cities consume much more power than the Koeberg power station can supply and rely heavily on its stability. The composite power must be supplied by the northern grid of South Africa. This makes the Western Cape's AC network relatively weaker than the northern regions especially during maintenance and refuel-scheduling of the reactors at Koeberg. An HVDC link between Cape Town and Koeberg would be highly beneficial to the existing AC network and provide fast and dynamic reactive power control and voltage stability control which would greatly improve the quality of supply and reliability of the southern and western grids. Since DC transmission losses are minimal in comparison to AC transmission, the reduced losses through a direct current undersea transmission to Cape Town also poses a small but consistent economic benefit in reduced transmission losses.

The worldwide attention on High Voltage Direct Current (HVDC) networks has grown in recent years with the advent of the reduction in the infrastructure cost, increased reliability and technological advancement of power electronic switching devices used in HVDC systems [1]-[5]. This has resulted in many new HVDC projects being successfully completed and new projects currently under construction [6][8]. New transmission systems favour HVDC over HVAC because of its increased and fast-responsive power system stability control [1]. There are also special cases of land transmission, roughly above 700 km, and undersea transmission, roughly above 30

km, where HVDC transmission is favourable, both from an economic and stability standpoint.

In HVDC systems, the AC signal of a generator or transmission line is converted to DC or back to AC using either a Current Source Converter (CSC) or a Voltage Source Converter (VSC) [1][3]. There are three popular options for converter types, namely: the Line Commutated Converter (LCC) and Capacitor Commutated Converter (CCC), for CSCs otherwise referred to as classical HVDC, as well as Self-commutated Converters (SCC), for VSCs. Since the advent of IGBTs, VSCs have begun to dominate the HVDC market for various reasons which will be discussed. However, CSCs using thyristors for switching, still have favourable application with very high power transmission applications above 2000 MW [1][3].

The technological advancement in VSCs has progressed with the most advanced system being the multi-terminal VSC HVDC (MTDC) system [1][2]. This grid system offers solutions to DC fault stability and harmonic filtering among other stability control benefits over isolated VSC HVDC. It provides interactive control of power and voltage between terminals. It also presents niche capabilities for transmitting power at various voltage levels and frequencies to different consumers [1]. HVDC is also ideal for applications of connecting asynchronous AC networks. VSC HVDC is particularly useful when connected to weak AC systems or island networks [9].

II. SPECIFICATION

It is important, when designing an HVDC link between Koeberg power station and Cape Town, to consider the various specifications surrounding Koeberg Nuclear Power Station (KNPS) and the City of Cape Town. In particular for KNPS, it is important to know its power output, the AC transmission voltage level leaving KNPS and its maintenance schedule. For the City of Cape Town, the voltage transmission level that it would preferably receive, its yearly power demand trend and the vicinity that would be most ideal for the distribution side converter of the DC link. It is imperative to clearly outline the requirements of the design to define an objective to be maintained throughout the design process.

TABLE I. KOEBURG POWER PLANT SPECIFICATION

Description	Value	Description	Value
Two generating units @ 900 MW	1800 kW	Uranium Fuel	U238 (U235 4% enriched)
Design efficiency @ rated TMCR	32.4 %	Voltage transmission level	400kV
Average availability over 3 years	83.1 %	Turbine	Pressurized water
Average production over 3 years	13668 GWh		

A. Koeberg Nuclear Power Station

Table I gives the specification for the Koeberg power station [10]-[12]. Each generator unit of the power station must undergo refuelling which can take up to 6 weeks each.

B. City of Cape Town

Cape Town is located approximately 27.4 km from Koeberg power station and the direct distance from Koeberg to Cape Town is 24.2 km, see Fig. 1. The nearest main substation is located at Green Point near the shore [13]. This location would seem the best option for interconnection to the AC power grid of Cape Town. The average power demand of the City of Cape Town is approximately consistent at just above 1900 MW until the end of 2017 [12]. According to the above information the following is assumed: distance to Cape Town: 27.4 km; distance Underwater: 24.2 km; and power demand: 1900 MW.Units.

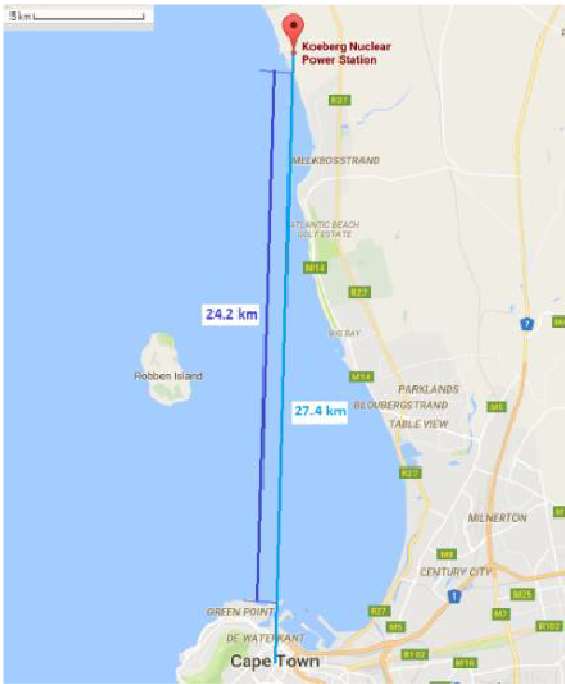


Fig. 1. Distance between Koeberg Nuclear Power Plant and Cape Town (courtesy Google Maps).

C. HVDC Cables

The selection of the type of HVDC cable is significant. The selection of the subsea cable and factors affecting it, such as depths, sea waves, tidal currents, and sea temperature will affect performance and space constraints prevents further discussion of these. However, extruded XLPE cables could be selected for this study and these can transmit the required 1800 MW at or below 525 kV, so this type of cable is ideal for our application. According to ABB [14], their 525 kV cable would

be capable of transmitting power of up to 2.6 GW for subsea applications. Therefore, this would mean the cable is capable of transmitting, at maximum conductor core diameter, $I = P/V = 2.6 \text{ GW}/525 \text{ kV} = 4952.4 \text{ A}$. If we are to assume that Koeberg transmits at maximum real power the current can be calculated from Table I where $I = P/V = 1.8 \text{ GW}/525 \text{ kV} = 3428.6 \text{ A}$. This would mean a maximum operational current that is well within the capacity rating of the conductor. However, according to the available-for-purchase technology as presented by ABB [15], their current maximum capacity XLPE copper single-core cable with a cross-sectional area of 1000 mm², is in a voltage range of 100 kV – 420 kV. If multiple cables are widely spaced their maximum current rating is 1290 A. If an operational voltage level of 420 kV is assumed, the current through one conductor at maximum power capacity of 1800 MW would be $1.8 \text{ GW}/420 \text{ kV} = 4285.7 \text{ A}$. Therefore, we can calculate the required number of cables = $4285.7 \text{ A}/1290 \text{ A} = 3.32 = 4$ conductors. This eliminates the practical implementation of a single converter station pair implemented in a bipolar configuration with two cables since the current flow through the cables would exceed the current rating of the highest voltage level transmission cable available. Two bipolar converter stations would have to be used with XLPE cable. If the proposed 525 kV cable technology does not exist and a single converter station pair configured in bipolar design is to be adopted, the alternative option would be to use MI cables. [16] states the main insulation of Mass Impregnated (MI) cables is Kraft paper which has a high density of equivalently 1000 kg/cm². This high density paper is impregnated with a high viscosity compound-based mineral oil which produces a high dielectric strength for cable insulation. Maximum operating voltages in recent years have been around 500 kV but new technology claims an operational voltage of 700 kV.

III. SELECTION OF CONVERTER

Here, a modular multi-level converter is used. However, in HVDC this is a relatively recent development. The options are discussed below.

Line commutated converters (LCCs) provide the necessary tools for transmitting high voltage direct current. They are simpler to design and implement and there is a history of successful implementation since the 1930s. The technology has been refined and improved over the last 80 years. However, LCCs present additional costs in the form of line compensation and synchronous condensers (for the application of a DC link on the isolated western South African grid which is weaker than its northern counterpart to which it relies on for long distance power transmission supplement). Synchronous condensers may be necessary to avoid commutation failure of the transistors. Line compensation is necessary to provide the converter active thyristors with the reactive power which they consume. The AC yard with filters and shunt compensation for

the LCC can take up as much as three quarters of the total area requirements of the converter station thus greatly influencing yard size. LCCs also do not provide any dynamic control in terms of voltage stability and reactive power. They also do not provide any additional benefits to power system stability in general as well as with special consideration of possible future connections of micro generators and implementations of smart grid technology.

Voltage source converters (VSCs) are self-commutated and therefore do not require synchronous condensers. They also do not require line compensation and instead can provide reactive power for the power network. They require capacitors within the converter structure which adds cost but this cost is less than that posed by LCC synchronous condensers and line commutation. VSCs can offer dynamic control for voltage regulation and independent active and reactive power control for power system stability. The energy stored in the capacitors of the converter can be used for generator black-start. Power can be transferred in both directions without the need for voltage polarity reversal which could be useful in the event that Koeberg power station experiences a frequency drop due to a power imbalance between generation and load and the plant requires power to avoid system destabilisation. VSCs can provide power system stability control if possible future micro generators are connected to the power system (which can threaten this stability). They offer greater dynamic control which is significant in light of smart grid technology development and possible future implementation.

Since the voltage source converter can provide greater control, and can dynamically influence the AC network it connects to, the VSC seems are far better choice than the LCC. This is especially relevant since the main LCC application today is for transmission of very high power over long distances which are not requirements for this project. Both types can transmit the required power of 2×900 MW and convert the voltage at the distribution end to a voltage level that can be supplied to a transformer to transform into a 400 kV voltage level. Additionally, both types can reduce transmission losses which adds to the attraction of an HVDC link in the form of a long term cost benefit. The VSC capital cost is less than the LCC for an increased number of benefits for the AC power network. The VSC also requires a smaller area for the station yard than a LCC.

Multi-terminal HVDC (MTDC) converter configurations provide for the interconnection of multiple AC networks. As stated in [4] the basic transmission circuit consists of a bipolar two-wire HVDC system with converters connected pole-to-pole. DC capacitors are used to provide a stiff DC voltage source. The DC capacitors are grounded at their electrical centre point to establish the earth reference potential for the transmission system. There is no earth return operation. The converters are coupled to the AC system through AC phase reactors and power transformers. Unlike most conventional HVDC systems, harmonic filters are located between the phase reactors and power transformers. Therefore, the transformers are not exposed to DC voltage stresses or harmonic loading, allowing the use of ordinary power transformers [4]. According to [17], the MTDC system power flow problem can be solved using the techniques of DC slack bus control, voltage margin

control and distributed voltage droop control. One terminal is operated as a slack bus for the DC network which is the equivalent of the largest synchronous generator in an AC network being used as the slack bus. The converter terminal in MTDC systems that is used as the slack bus may need to be overrated and connected to a strong AC network. Voltage margin control offers an improvement in constant dc voltage control. Voltage droop control allows for multiple converter terminals to contribute to maintaining the DC voltage.

The modular multilevel converter (MMC) is a converter topology that presents the switching IGBTs in cell-like modules of specific configuration. The topology has a scalable construction [17]. Fig. 2 shows the MMC construction and cell configuration.

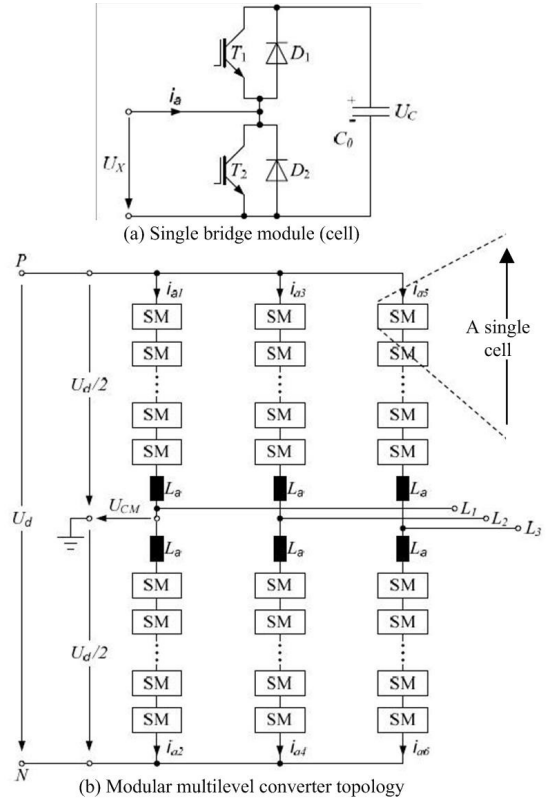


Fig. 2. MMC construction and cell configuration [17].

Here the development of the MMC model will be discussed in detail including the control. An example 5 level converter will be first described and then this is extended to a 200 level MMC with the Ansys Simplorer software. This will show the module details. Here we will put forward the final 200 level converter with initial results.

IV. FULL SYSTEM

To develop the system then a 5-level converter was first to test the system and then a full system model developed.

A. 5-level model half bridge and full 3-phase inverter

The initial half-bridge MMC has the parameters shown in Table II. This was developed as illustrated in Fig. 3. This was used as a testing circuit and the output waveform across the load resistance R is shown in Fig 4(a). One of the problems

with the simulations is the balancing of the capacitors as illustrated in Fig 4(b). C_1 and C_3 increase to very high voltage and tend towards a constant value whilst C_2 and C_4 tend to zero. This is due to a voltage imbalance in the capacitors.

TABLE II. VALUES USED IN 5-LEVEL MMC SIMULATION

V_{dc}	5 kV
R	10 ohms
L_3	23.8 mH
$F_{carrier}$	1.35 kHz
m	0.9
$L_1 = L_2$	1 mH
C_n for $n = 1, 2, 3, 4$	3 mF

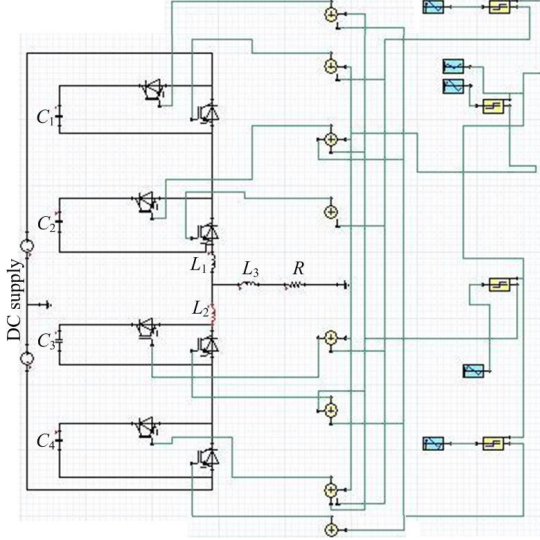


Fig. 3. 5-level half bridge.

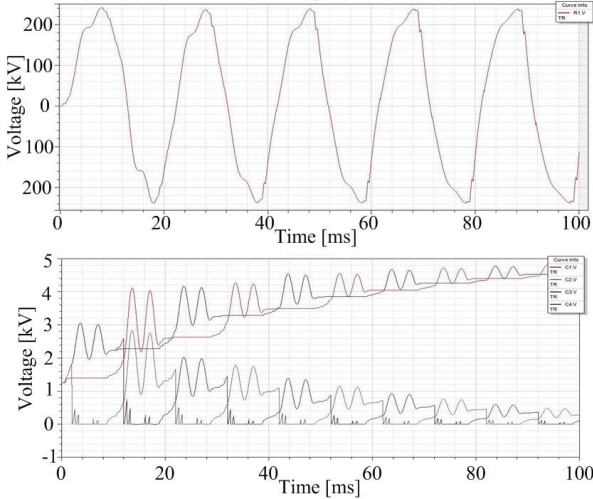


Fig. 4. Initial 5-level half bridge simulation results. (a) Output voltage and (b) capacitor imbalance.

This imbalance is caused by the modulation control's linearity coupled with the simulated tolerances of the components as well as the in-rush current due to only one modulating signal. In order to rectify this issue, a PI controller is needed in the eventual closed-loop control that can reduce the steady state error and drive the capacitors towards being balanced. It would also benefit this system to have two independent modulating signals, one for each converter arm,

which are 180 degrees out of phase with one another, to eliminate the in-rush current on the zero crossing. This voltage imbalance is illustrated in Fig. 4(a) where the voltage waveform is distorted. As suggested in by the title the aim of this paper is to give experiences in simulation of MMC and balancing the capacitors is a key issue. The system was extended to a full three-phase system, but space constraints prevents further reporting of this system.

B. Full system

The full system consists of a transformer connected to the AC supply from the power grid which steps down the high line voltage (assumed 400 kV) to an appropriate value for the rectifier. This value was chosen to 160 kV peak in order to produce a 320 kV HVDC. The power from the AC system is then transmitted through the submarine cables to the inverter station at Greenpoint, Cape Town. The inverter, through gated control SPWM and a capacitor voltage control system of a 200-level three-phase inverter system, converts the HVDC to HVAC. This HVAC is then supplied to the distribution side transformer which steps down the voltage to an appropriate distribution voltage level or the voltage level at which the nearest substation would desire. The voltage preliminarily chosen for the secondary of the distribution side transformer is 12 kV. Through the simulations it was noticed that the voltage delivered to the AC side of the inverter was approximately 120 kV so a transformer with ratio of 0.1 would be required. The full system block diagram is shown in Fig. 5(a). An attempt was made to model this in Simpler as illustrated in Fig. 5(b) but it was found that this system reached and exceeded the capacity of the software and computer used. The system complexity and numerous cells reduced the software and CPU (3.40 GHz, 8MB, 64-bit) speed capabilities such that the copying of a single-phase arm caused the program to freeze for 75 minutes at a time. However, an inverter was possible and the next section discusses the 200 level inverter in detail.

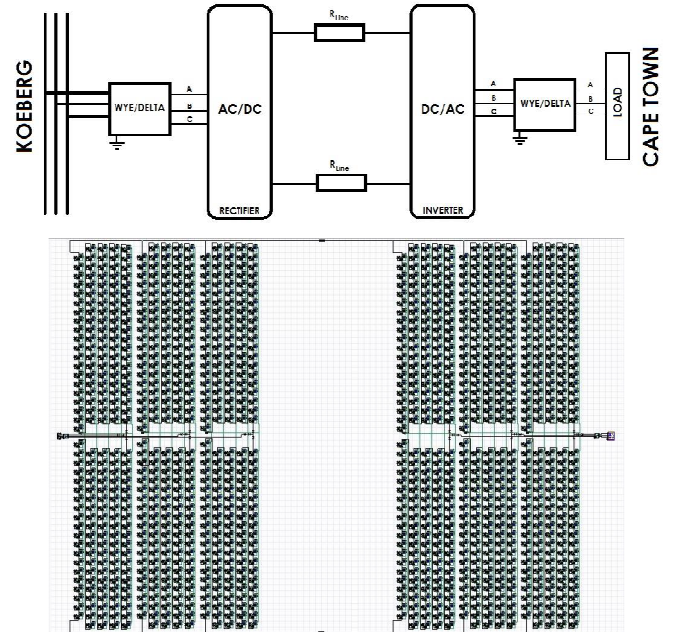


Fig. 5. Full system. (a) Block diagram and (b) Simpler representation.

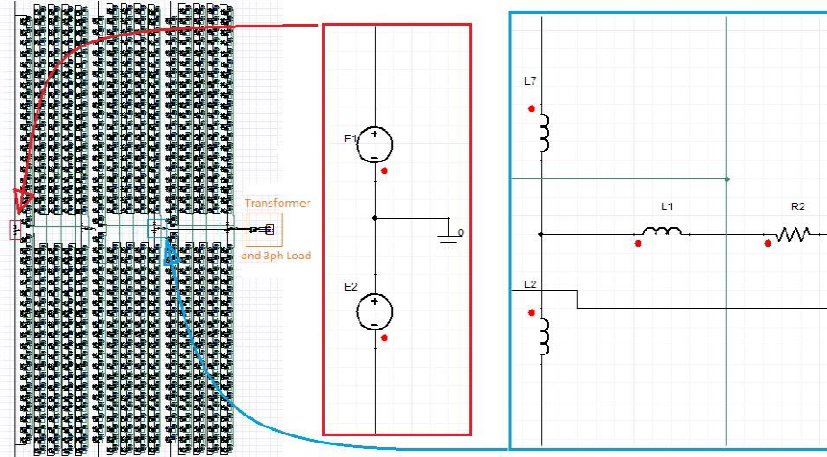


Fig. 6. HVDC link system designed in ANSYS Simplorer – assumed perfect rectification so only inverter studied.

V. 200 LEVEL CONVERTER

The system chosen, with a voltage source converter, requires the construction of the rectifier and inverter stations using IGBTs, diodes and capacitors. Inductors implemented in series help smooth the switching level voltage output waveform. A load inductor and resistor was included to complete the test system and the output voltage was measured across the load resistor. A complete system was constructed in ANSYS Simplorer. However, computer limitations meant that rather than modelling both the rectifier and inverter then ideal rectification was assumed and only the inverter was studied as shown in Fig. 6. Each module was modeled at component level as illustrated in Fig. 5 which gives the converter cell circuit and the gate control.

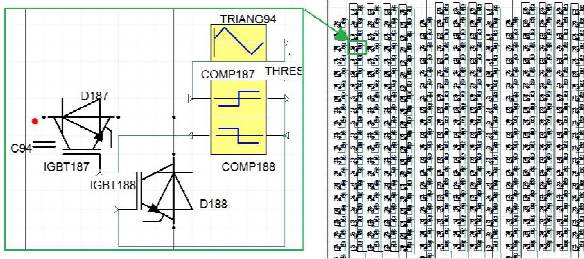


Fig. 7. Converter cell circuit diagram with gate control.

A. Gate control

The control of the capacitor voltages in the cells of the MMC is essential for stable operation of the DC link. In [18], the control of the capacitor voltage is simplified by obtaining the cell capacitance reference voltage by determining two component values using closed-loop control. The sum of these two values and the arm voltage to ground yields the reference voltage which is used to realise the capacitor balancing control. Observe Fig. 8(a) obtained from [18] which represents the energy averaging control loops. This control system relies on the average energy versus actual energy balance in the cells. If there is an imbalance in this energy then the system outputs represent this imbalance error. The function of the PI control blocks is to fine tune the second order behaviour and response of the system in each loop: Voltage Outer Loop and Current Inner Loop; and remove the steady state error to an insignificantly low value. An additional level of control is

necessary for each cell. This control deals with the voltage across each cell capacitor relative to the capacitor reference voltage. This is observed in the individual-balancing control from [18] and shown in Fig. 8(b).

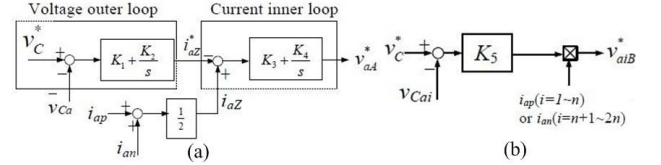


Fig. 8. Current balancing. (a) Energy average control system with PI control and (b) individual-balancing control system with proportional control.

B. Simulation of system

The inverter was designed with 100 cells per arm and not the calculated 101 for simplicity of simulation values. Each individual triangular wave generator required a unique dc offset value that was fairly accurate in terms of decimal point (up to 6 decimal points since up to 1 us step) so using 100 cells per arm conveniently set the maximum decimal places to 3. The amplitude A and offset, ρ_N , for each triangular carrier were calculated using

$$\rho_N = \pm \frac{2N-1}{2k} \quad (1)$$

where $N = 1, 2, 3, \dots, k$ and k is the number of cells per arm. The negative value is taken when dealing with the negative arm carriers. The sinusoidal reference voltage magnitude or the modulation index was varied between 0.9 and 1.

For the system in Fig. 6, each DC source supplies 320 kV. The series inductors L_1 and $L_2 = 0.1$ mH, $L_1 = 23.8$ mH and $R_2 = 10 \Omega$. In Fig. 7, the THRES input into COMP187 (comparator) represents the threshold value of the comparator and is the reference sinusoidal signal. COMP188 acts to enforce the complementary nature of the IGBTs. TRIANG94 is the triangular wave generator. The cell capacitance is 3 F. The triangular wave generator amplitude is: $A = 1/2k$. The offset ρ_N in (1) is unique for each triangular wave and must be individually set for all 600 cells in the inverter.

C. Results

The initial output phase voltages and currents for the inverter are shown in Fig. 9. This still has not reach steady state

and the system is under load. The current waveforms track the voltages since the load is resistive. As can be seen, with 200 levels, the waveforms appear good. However, it can observe that the voltage is about 120 kV peak. The voltage has some instability due to the absence of capacitor control for each capacitor. It is important for any modular multilevel converter to possess this control and that this control must function correctly and without error. Due to this imbalance, a DC component develops in the voltage output. The imbalance of the capacitor voltages can be observed in Fig. 9. This shows the immediate response of the capacitors to the gated switching where some capacitors receive higher charge over time than they release into the system whilst others receive and release the opposite. The output voltage will tend to converge towards the DC component and the system will fail due to the disparity in the on and off times between the cells. This can be resolved by use of a capacitor voltage control as previously discussed.

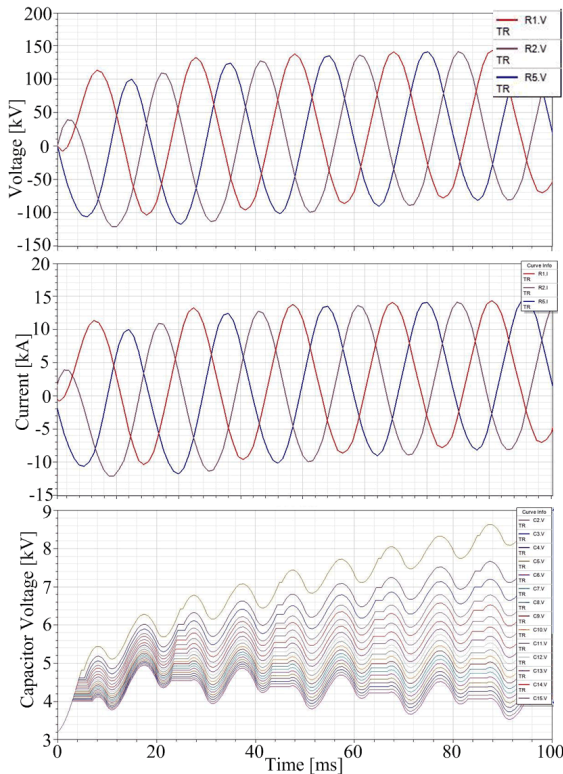


Fig. 9. Output voltage and current of each phase in the 3-phase 200-level inverter (time step 1 ms) and capacitor voltages.

VI. CONCLUSIONS

Multiple types of HVDC converter system design options were considered. The HVDC system selected to simulate a DC link between Koeberg and Cape was the modular multilevel converter system topology. This used a bipolar design utilising IGBTs and amplitude sinusoidal pulse width modulation control to modulate the gates of the IGBTs. The cable type selected was XLPE cable. The distance undersea was determined to be approximately 24 km from Koeberg's to Cape Town's shoreline. Simulations were completed. These illustrated specific control issues with the static discrete control as well as with the in-rush current from having a single reference signal when using SPWM for low level test designs.

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